Using the Law of Conservation of Energy to solve problems

Student worksheet

Use $g = 9.8\ \text{m s}^{-2}$ throughout unless stated otherwise.

Objectives

- Identify the main energy transfers in a physical event.
- State and explain the principle of conservation of energy.
- Solve problems by comparing the total energy before and after an event.
- Use K, ΔU_g, U_g, U_s and ΔU_int with correct AP-style notation.
- Compare force-and-motion methods with energy methods and choose an efficient approach.

Core principle

Energy cannot be created or destroyed. For a suitably chosen system, total energy before an event equals total energy after the event.

Total energy before = Total energy after

Quantitative AP energy quantities used here: K, ΔU_g, U_g, U_s, and ΔU_{int}. We do not use the work-energy theorem in this lesson.

Task 1 · Identify the main energy transfer

1. A small, dense ball falls through the air.

Energy type 1: _____ → Energy type 2: _____

2. A car with its engine switched off coasts uphill and slows down.

Energy type 1: _____ → Energy type 2: _____

3. The school lift descends at steady speed.

Energy type 1: _____ → Energy type 2: _____

4. A compressed spring launches a cart on a horizontal track.

Energy type 1: _____ → Energy type 2: _____

5. A bicycle brakes to rest on level ground.

Energy type 1: _____ → Energy type 2: _____

6. A bouncy ball strikes the floor and compresses before rebounding.

Energy type 1: _____ → Energy type 2: _____

GREENSICK bank: Gravitational · Radiation · Elastic · Electrostatic · Nuclear · Sound · Internal · Chemical · Kinetic

Task 2 · Simple conservation of energy

A. A 100 W lightbulb converts 100 J of electrical energy each second. If 75 J becomes internal energy each second, what other energy change can be deduced?

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B. A ball loses 250 J of gravitational potential energy and gains 200 J of kinetic energy. Suggest what else has occurred.

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C. A stretched elastic band is released vertically upward. At the highest point the object has gained 25 J of gravitational potential energy. Neglect losses. What can you deduce about the elastic potential energy stored initially?

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Task 3 · Gravitational and kinetic energy

A 28 g ball falls from rest from a table 78 cm high.

Use: $K = \frac{1}{2}mv^2$ and $\Delta U_g = mg\Delta y$. Convert to SI units first.

(a) Determine Δy, including sign.

(b) Determine ΔU_g, including sign.

(c) Explain what the sign means.

(d) Use a kinematics equation to determine the speed just before impact.

(e) Determine K just before impact.

(f) Has the ball gained or lost kinetic energy? Compare the numerical change with ΔU_g.

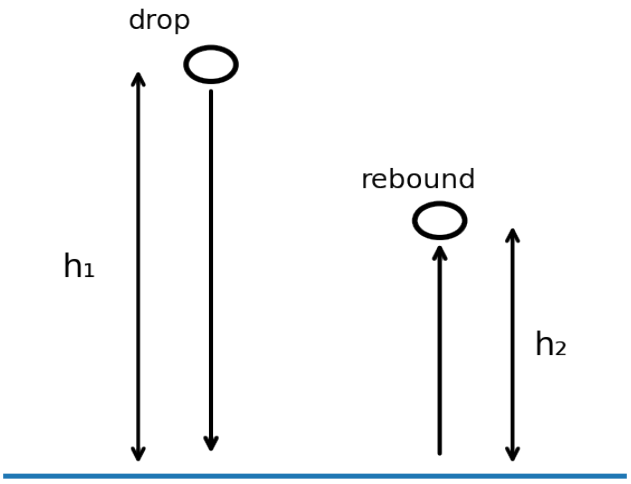
Task 4 · Before-and-after energy accounting

For each later problem, begin by writing a before/after statement and cross out terms that are zero or unchanged.

$$\mathbf{K_i + U_{g,i} + U_{s,i} + U_{int,i} = K_f + U_{g,f} + U_{s,f} + U_{int,f}}$$

For a ball dropped from rest with negligible drag, write the reduced equation:

Task 5 · Practical: bouncing ball



Equipment: bouncy ball, mass balance, metre stick, stopwatch.

Method: Measure mass m and drop height h_1. Release without pushing. Measure first rebound height h_2. Repeat at least five times and average. Optional: time between first and second impacts to estimate $h_2 \approx g(\Delta t)^2/8$.

Record mass: _____ kg Drop height h_1: _____ m

Rebound heights (m): 1 _____ 2 _____ 3 _____ 4 _____ 5 _____ mean _____

Calculate $E_{before} \approx mgh_1$.

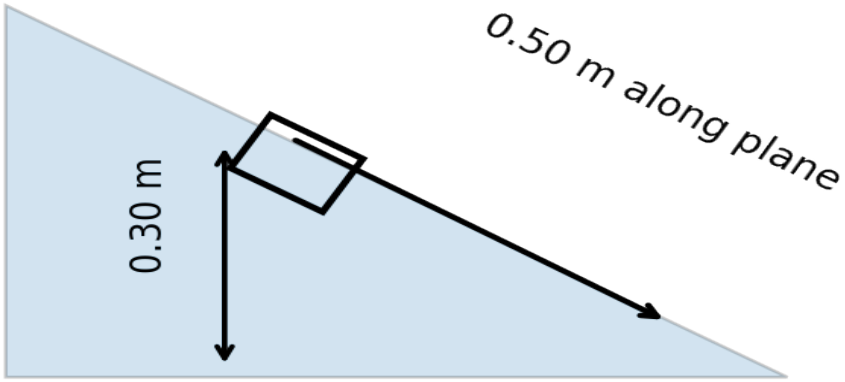
Calculate $E_{rebound} \approx mgh_2$.

Estimate mechanical energy not recovered: $mg(h_1 - h_2)$, and percentage recovered $100\ h_2/h_1$.

Discuss where the missing mechanical energy went and the main measurement uncertainty.

Task 6 · Comparing Force & Motion with Energy

A. A 1.2 kg block starts from rest and moves 0.50 m down a frictionless incline while dropping vertically by 0.30 m.



(a) Determine the incline angle.

(b) Determine the component of weight parallel to the plane.

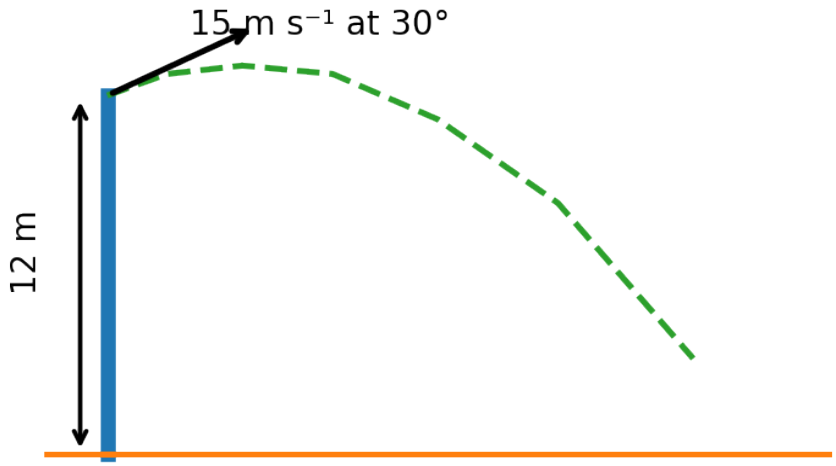
(c) Determine acceleration.

(d) Use kinematics to determine speed after 0.50 m.

(e) Determine ΔU_g.

(f) Determine K at the same instant and compare with part (e).

Task 6B · Projectile: compare motion and energy



A 0.25 kg ball is launched at 15 m s^{-1}, 30° above horizontal, from a building 12 m high. Air resistance is negligible. “Initial” = just after launch; “top” = maximum height.

(a) Find $v_{x,0}$.

(b) Find $v_{y,0}$.

(c) Use kinematics to find the rise above the roof.

(d) Find initial K.

(e) At the top, find v_x.

(f) At the top, find the speed.

(g) Find K at the top.

(h) Find ΔK from launch to top.

(i) Find ΔU_g from launch to top.

(j) Find ΔU_g from roof to ground.

(k) Hence find the change in K from launch to ground.

(l) Find impact speed.

Task 7 · Exploring gravitational potential energy further

Near Earth: $\Delta U_g = mg\Delta y$. For large distances: $U_g = -Gm_1m_2/r$, with zero at infinite separation.

Use $G = 6.67 \times 10^{-11}$ N m² kg^{-2}, $M_E = 5.98 \times 10^{24}$ kg, $R_E = 6380$ km, and $m = 65$ kg.

(a)(i) Find U_g at Earth's surface.

(a)(ii) Find U_g at $2R_E$.

(a)(iii) Find the increase in U_g.

(a)(iv) Find the energy required to escape to infinity.

(b) Use $\Delta U_g = mg\Delta y$ for a rise of 6380 km. Comment on why this answer is not valid for such a large height change.

Task 8 · Dropping a ball with drag

A 520 g ball falls from rest through 14 m.

(a) Determine the magnitude of its decrease in gravitational potential energy.

(b) If all of this becomes K, determine K.

(c) Hence determine speed.

(d) Instead, half the energy becomes internal energy. Determine final speed.

(e) Assuming constant acceleration, determine acceleration.

(f) Hence determine the air-resistance force using force analysis.

Task 9 · Generalising by deriving

(a) An object of mass M starts from rest and falls through vertical height H. By considering energy conversion $U_g \rightarrow K$, derive an expression for final speed v.

(b) The same object begins with downward speed v_0. Derive v after it falls through H. Comment on the result.

Task 10 · Projectiles

1. A 2.5 kg object is projected horizontally at 5.0 m s^{-1} from a building 12 m high. Ignore air resistance.

(a) Find initial K.

(b) Find the decrease in U_g and hence the increase in K. State the assumption.

(c) Find final K and impact speed.

2. Generalise: mass M, initial horizontal speed v_0, height H. Derive impact speed.

3. A 0.500 kg object is projected from ground level at 12 m s^{-1}, 60° above horizontal.

(a) Initial K.

(b) Horizontal velocity component.

(c) How does this component change in flight?

(d) Vertical velocity at highest point.

(e) K at highest point.

(f) K lost and U_g gained. State the assumption.

(g) Maximum height.

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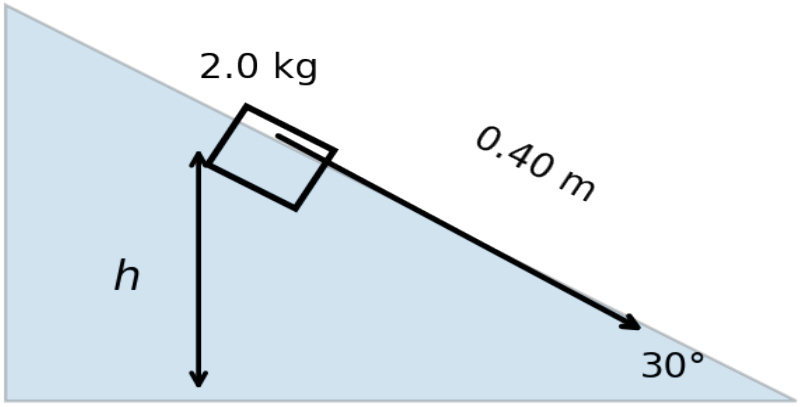
Which method do you prefer here: energy or uniformly accelerated motion? Explain.

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Task 11 · Inclined planes



A 2.0 kg block starts from rest on a frictionless plane 30° above horizontal and moves 0.40 m down the slope.

A. Force & motion

(a) Find the net force down the plane.

(b) Find acceleration.

(c) Find speed after 0.40 m.

B. Conservation of energy

(a) Find vertical drop h.

(b) Find decrease in U_g and final K.

(c) Find final speed.

C. Generalise: mass M, angle θ, distance d.

(a) h in terms of d and θ.

(b) gravitational-energy decrease in terms of M, g, d, θ.

(c) final K and speed.

(d) What does this show about mass and the motion of a frictionless block?

Task 12 · Challenge algebra: rough inclined plane

A block of mass m starts from rest and slides down a rough plane inclined at θ. Its vertical height drop is h and the coefficient of kinetic friction is μ. The acceleration is down the plane. Derive expressions using only m, θ, h, μ and g.

(a) distance d moved along plane.

(b) kinetic-friction force.

(c) net force down plane.

(d) acceleration.

(e) final speed.

(f) final K.

(g) ΔU_g.

(h) ΔU_{int}, using conservation of energy.

Questions I still have